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The effect of high pH on ion balance, nitrogen excretion and behaviour in freshwater fish from an eutrophic lake: A laboratory and field study

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Abstract

Slapton Ley is a freshwater hyper-eutrophic lake of two basins connected by a narrow channel. One part of the lake experiences summer blooms of cyanobacteria and poor water quality, including elevated water pH (maximum pH recorded = 10.54), the other part is shaded by reed beds, and remains clear and neutral. This study used laboratory and field physiological measurements together with radio-tracking to investigate the potential impacts of alkaline pH on the physiology and behaviour of fish from Slapton Ley. Exposure of perch (*Perca fluviatilis*) from Slapton Ley to pH 9.50 water in the laboratory caused an immediate inhibition of sodium uptake and ammonia excretion to 34 and 32% of control levels, respectively. Net sodium balance recovered by day 3 of exposure whereas ammonia excretion only partially recovered to 60–70% of the control value from 8 h onwards. Urea excretion did not increase as a result of high pH exposure. Fish from the alkaline part of the lake (pH 9.90) had almost three-fold greater plasma ammonia compared to fish from neutral waters, indicating a pronounced disruption of ammonia excretion in the field. There was no significant disturbance to plasma sodium, chloride or total protein in fish sampled from the alkaline water of Slapton Ley. The radio-tracking provided no evidence of adult perch and pike (*Esox lucius*) trying to seek refuge from the alkaline conditions, despite having access to adjacent parts of the lake with neutral pH. It seems likely that there are advantages (e.g. better foraging, less predation) of withstanding the high pH conditions that outweigh the benefit of moving into more pH neutral parts of the lake.

Keywords: Eutrophic lake; Alkaline pH; Ion regulation; Nitrogen excretion; Perch (*Perca fluviatilis*); Pike (*Esox lucius*); Roach (*Rutilus rutilus*); Radio-tracking

1. Introduction

Information on the effects of long-term exposure to high pH water on ion balance and ammonia excretion in freshwater fish is mainly limited to salmonids such

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as the rainbow trout (*Oncorhynchus mykiss*). In these fish alkaline water (pH 9.5) causes immediate reduction in the branchial uptake and plasma levels of Na^+ and Cl^-, with no effect on branchial diffusive efflux (Wilkie and Wood, 1991, 1994; Wilkie et al., 1996, 1999; Laurent et al., 2000). Long-term survival at high pH requires regulation of internal electrolyte balance by restoration of branchial ion movements. Cl^- uptake recovers fully whilst Na^+ uptake only partially recovers (Wilkie and Wood, 1994; Wilkie et al., 1996, 1999; Laurent et al., 2000). Net Na^+ balance is partially restored by a reduction in Na^+ efflux rather than recovery of Na^+ influx, which may be a favourable mechanism allowing fish to retain more H^+ and thus help to offset alkalosis, which is a symptom of exposure to alkaline water (Goss et al., 1992; Yesaki and Iwama, 1992; Wilkie et al., 1993). Alkaline water also causes immediate, dramatic inhibition of ammonia excretion and subsequent increase in plasma ammonia, which can be potentially lethal (Wright and Wood, 1985; Wilkie and Wood, 1991, 1994, 1995; Yesaki and Iwama, 1992; Wilkie et al., 1996; Wilson et al., 1998; Laurent et al., 2000). The rate of ammonia excretion in freshwater fish is thought to be determined by the magnitude of the NH_3 partial pressure gradient across the gills (Wilson et al., 1994; Salama et al., 1999). Inhibition of ammonia excretion is attributed to disruption of this gradient, due to a reduction in H^+ ions available to trap NH_3 as NH_4^+ in the gill boundary layer (Wright et al., 1989). In rainbow trout full or partial recovery of ammonia excretion has been observed within 24–48 h of high pH exposure (Wilkie and Wood, 1991, 1994, 1995; Wilkie et al., 1996; Wilson et al., 1998; Laurent et al., 2000).

The movement of an animal in its environment is a response to endogenous and external factors enabling it to feed, compete, reproduce and escape predators and unfavourable environments (Craig, 1987). Wildlife telemetry enables the behaviour of free-ranging animals to be studied, without the need for repeated intervention after tagging (Priede and Swift, 1992; Lucas and Baras, 2000; Jepsen et al., 2002). Although numerous studies have investigated fish behaviour in relation to a wide variety of environmental factors, information on the effect that changes in water pH have on fish behaviour is sparse. One laboratory study found that three species of fish avoided pH levels greater than pH 9.5, except when combined with supersaturated oxygen levels (Serfay and Harrell, 1993). At the other extreme, Carline et al. (1992) radio-tracked brook trout (*Salvelinus fontinalis*) in a stream subject to toxic acid flushes, during which trout retreated downstream; although this response may have been due to elevated aluminium as much as to low pH.

The study site, Slapton Ley, is a freshwater lake that is divided into two basins, the Higher and Lower Leys, joined by a narrow water channel (Fig. 1). The Higher Ley (39 ha) is mainly reed swamp whereas the majority of the Lower Ley (77 ha) is open water fringed with reed beds (Van Vlymen, 1980; Morey, 1976). Since 1950 Slapton Ley has become increasingly eutrophic, associated with intensification of agriculture and inputs from sewage treatment works (O'Sullivan et al., 1991; O'Sullivan, 1994; Johnes and Heathwaite, 1997), and is currently regarded as hyper-eutrophic (Scott, 2003). Cyanobacterial blooms (*Microcystis* and *Anabaena* spp.) in the Lower Ley are now a common phenomenon, along with extreme seasonal fluctuations in the pH (Fig. 1) (O'Sullivan, 1994; Scott, 2003). The Higher Ley on the other hand, which is shaded by reed beds, remains clear of cyanobacteria blooms and near neutral pH all year round (Scott, 2003).

Little is known about the physiological and behavioural responses of non-salmonid fish to alkaline conditions. The aim of this study was to investigate the potential impacts of chronic exposure to high pH on the physiology and behaviour of fish from Slapton Ley and the implications for survival of wild fish under such conditions. In particular, we asked how chronic high pH exposure affects the ion balance and nitrogen excretion of Slapton fish and if wild fish seek refuge in more neutral waters during high pH episodes in parts of this hyper-eutrophic lake?

2. Methods

2.1. Laboratory experiments—chronic effects of high pH

2.1.1. Experimental animals

Two groups of perch were caught from Slapton Ley by electric fishing using a pulsed DC unit (Safari Research Surveyor; peak power output = 2.8 kW). The first group of fish (mean mass ± S.E.M. = 3.76 ± 0.2 g, $n = 20$) were used to measure Na^+ balance and ammonia excretion, and the second group (36.8 ± 1.7 g,

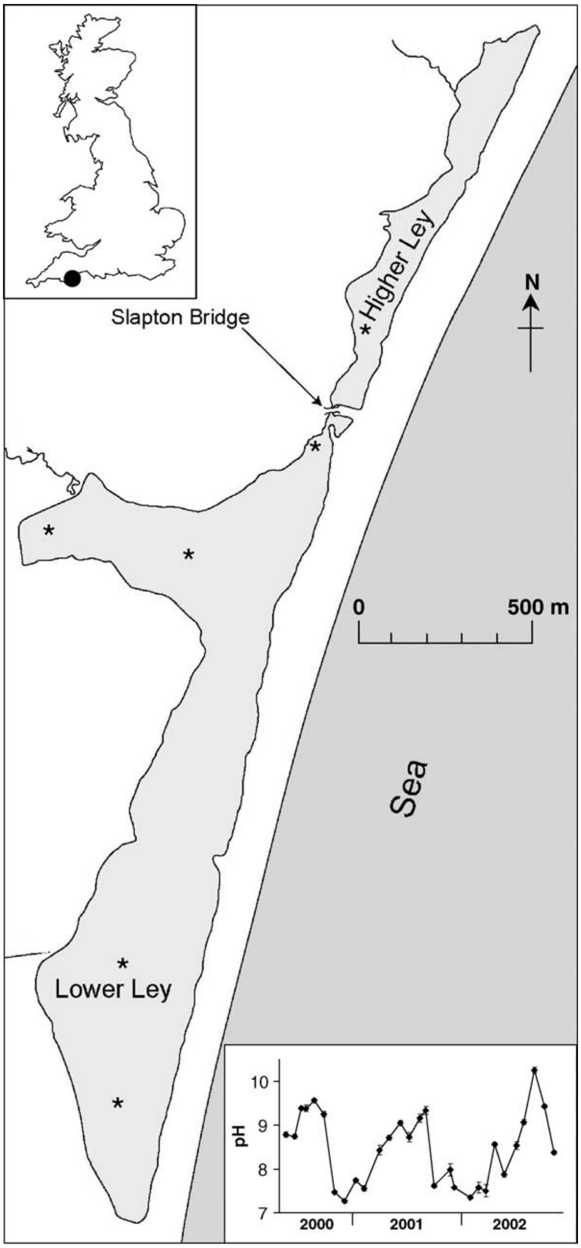

Fig. 1. Map of Slapton Ley. Inset in top left corner shows position within the UK. Asterisks show the sites where water quality analysis was conducted during the radio-tracking study. Bottom right inset shows the average pH ± standard error of the Lower Ley between June 2000 and October 2002, means are from monthly measurements made at 17 sites around the lake (Scott, 2003).

$n = 12$) to measure urea excretion during exposure to chronic high pH. Fish were maintained in dechlorinated Exeter tap water (average composition: $[Na^+] = 411.8 \pm 15.3$, $[K^+] = 54.8 \pm 2.8$, $[Ca^{2+}] = 623.6 \pm 17.5$, $[Mg^{2+}] = 138.5 \pm 6.9$, titratable alkalinity = $968.5 \pm 42.4\ \mu mol\ l^{-1}$, pH = 7.46 ± 0.04, temperature = $18.1 \pm 0.2\ ^{\circ}C$) for approximately 1 month prior to experiments. Experimental animals were fed with commercial fish food and live chironomid larvae from in-house cultures. Photoperiod was 12:12 h light/dark.

2.1.2. *The flux systems*

Two identical flux systems (control and experimental) were set up side by side, each consisting of a recirculating volume of water (150 l) that fed 10 darkened, aerated, 125 ml flux chambers in which individual fish were placed. Water in the flux system was aerated and chilled to 18.0 °C (Grant RC 1400G chiller) and supplied to each flux chamber at a flow rate of 350 ml min^{-1}. During the experimental fluxes the pH of the water was raised using a pH controller (Hanna HI 8710E) and pH electrode (Russell KCET/11 electrode) to deliver 0.1 mol l^{-1} KOH via a solenoid valve into the reservoir tank to maintain the pH between 9.45 and 9.55. The pH of the water in the chambers was also checked independently with the use of a hand held pH meter (Hanna HI 8710) and electrode (Russell KCET/11). During fluxes, when the flow of water to the flux chambers was stopped, the pH within the chambers in the experimental system was maintained by manual addition of 0.1 mol l^{-1} KOH.

2.1.3. *Experimental protocol*

Food was withheld from fish during the week prior to an experiment in order to allow ammonia excretion rates to stabilise (Brett and Zala, 1975). Twenty-four hours prior to an experiment individual fish were transferred into flux chambers to allow metabolic and ionic fluxes to stabilise following the stress of handling. Each flux consisted of a period of time in which the flow to the chambers was stopped and the hole in the side of the chamber closed. At the beginning of the flux period $^{22}NaCl$ (NEN, NEZ081) was added to the chamber water at 0.074 KBq ml^{-1} to allow the measurement of unidirectional Na^+ fluxes. Following a 5 min mixing period an initial 15 ml water sample was removed, followed by a final water sample taken at the end of the flux period. The chamber water was then siphoned out and flushed five times in order to remove any radioactivity before reconnection to the recirculating system.

Individual fish were anaesthetised with tricaine methanesulphonate (MS-222; 100 mg l^{-1}), neutralised with KOH, prior to transfer to the flux chambers to allow the accurate measurement of length and mass. After 24 h of recovery a series of 1 h fluxes were conducted. Firstly a pre-exposure flux was completed at the normal pH of Exeter dechlorinated tap water (pH 7.46). Following flushing of the chambers, the pH of the experimental system was acutely raised to pH 9.50 by addition of 0.1 mol l^{-1} KOH. Once this pH was reached in all chambers the 0–1 h flux was started. A pH of 9.50 was maintained within the system for 1 week and hour long fluxes were conducted after 4, 8, 24, 48, 72 and 168 h (1 week). In between the 72 h and 1 week fluxes, fish were put together in groups of five individuals within four 6-l tanks receiving either control or experimental water from the same system as before. This was to avoid keeping the fish confined in a small chamber for longer than necessary. Fish were taken out of this larger tank and returned to the flux chambers 24 h prior to the start of the flux on day 7. At the end of the experiment fish were identified by mass and length. The water in both the systems was replaced every 24 h with fresh, dechlorinated Exeter tap water. The K^+ concentrations in the experimental system never exceeded 1.5 mmol l^{-1}, a value well below that thought to be toxic to the fish (Wilkie et al., 1993, 1996).

This experiment was repeated with a separate group of perch (six control and six experimental) in order to measure urea excretion. These were larger fish than previously used and, therefore, larger chamber volumes (630 ml) were required. Fish were not anaesthetised before the start of the experiment and remained within the experimental chambers for the duration of the experiment.

2.1.4. Analytical techniques and flux calculations

The water samples taken at the beginning and end of each flux period were analysed for $[Na^+]$, ^{22}Na, total ammonia ($[T_{Amm}]$) and urea. Ion concentrations (Ca^{2+}, Mg^{2+}, K^+ and Na^+) were measured using atomic absorption spectrophotometry (Pye Unicam SP9). For the radioactive samples of water $[Na^+]$ was measured by flame photometry (Corning 410) in a fume hood. Concentration of total ammonia $[T_{Amm}]$ in water samples was measured in triplicate using a micro-modification of the salicylate–hypochlorite assay (Verdouw et al., 1978), and the urea was assayed using the diacetyl monoxime method (Rahmatullah and Boyde, 1980) (Cecil 1010 Spectrophotometer). Triplicate 2 ml water samples were analysed for ^{22}Na activity, counted on a Packard Cobra auto-gamma counter (B5002). Titratable alkalinity was determined as described by McDonald and Wood (1981); water samples were titrated to a fixed end point (pH 4.0) with 0.02 mol l^{-1} HCl. Unidirectional and net Na^+ fluxes, ammonia and urea excretion rates were calculated as described by Wright and Wood (1985).

When the Na^+ flux values, ammonia and urea excretion rates were normally distributed, they were compared using one-way repeated measures ANOVA followed by Tukey's test to assess the statistical significance of the differences between means at the different time-points. $P < 0.05$ was accepted as indicating statistical difference. Normality was checked using the Kolmogorov–Smirnov test. When normality failed, values were compared using repeated measures ANOVA on ranks ($P \leq 0.05$) (Dunnett's method) with each fish as its own control.

2.2. Field sampling and physiological measurements

In order to investigate the physiological status of wild fish during an episode of high pH water, blood sampling was conducted on fish caught on-site at Slapton Ley. On 30 August 2002, fish were caught by electric fishing from the Lower and Higher Leys (high and neutral pH, respectively). Roach (mean mass = 55.9 ± 4.1 g, n = 9) and perch (mean mass = 45.5 ± 29.8 g, n = 5) were caught in the Lower Ley (pH 9.9, temperature 20.2 °C) and perch only (mean mass = 20.6 ± 2.8 g, n = 11) were caught in the Higher Ley (pH 7.56, temperature 18.3 °C). Fish were held in aerated containers of the same water they were captured in for 1–2 h before being sampled. Fish were given a blow to the head prior to blood sampling via caudal puncture with heparinised (Monoparin Na^+-heparin) syringes and needles, and the brain was destroyed. The blood samples were kept on ice for 20 min before being centrifuged at 9500 rpm for 5 min. Plasma was decanted and kept on dry ice, returned to the laboratory within 4 h and then kept frozen (−80 °C) until the analysis was conducted. Fish that were not sampled in the field were brought back and held in Exeter dechlorinated tap water, and were blood sampled (as above) 48 h after the initial field plasma samples had been taken. The only fish remaining for this 48-h sampling were perch from the Higher Ley (mean mass = 3.1 ± 0.2 g, n = 9) and roach from the Lower Ley (mean mass = 40.1 ± 4.2 g, n = 10).

Plasma ammonia was measured by enzymatic determination, using a micro-modification of the Sigma kit (171-UV). Protein, Cl^- and Na^+ were also measured where there was sufficient remaining plasma. Plasma protein was measured using the Lowry assay (Lowry et al., 1951; Miller, 1959). Samples were diluted 100-fold and bovine serum albumin (Sigma A-9647) solutions were used as standards. Cl^- was measured by the colourimetric mercuric thiocyanate/acidic ferric nitrate assay (Zall et al., 1956) and Na^+ was measured using atomic absorption spectrophotometry. Results of the plasma analysis were compared using one-way ANOVA ($P \leq 0.05$) with pairwise multiple comparison procedures (Tukey's test).

2.3. Radio-tracking study—movements of fish within Slapton Ley

It was apparent from a previous fyke netting study that fish do move between the Higher and Lower Leys in considerable numbers on occasion (Scott, 2003). Therefore, a radio-tracking study was devised to examine the fish movement between these areas in response to changes in water chemistry. Eighteen perch (mean length = 20.6 ± 0.5 cm, mass = 144.5 ± 10.3 g) and nine pike (mean length = 59.3 ± 3.1 cm, mass = 1.69 ± 0.23 kg), were captured by electric fishing, angling, seine netting or fyke netting from Slapton Ley over the period from 22 May to 31 June 2002. The start of this period was chosen to avoid, and allow sufficient time for recovery of condition following spawning (Craig, 1974; Bregazzi, 1978; Bregazzi and Kennedy, 1980). Prior to tagging with radio-transmitters fish were held in keep nets within the lake for no longer than 12 h.

2.3.1. Radio-transmitters and implantation

Four sizes of radio-transmitters (Biotrack Ltd., UK) were used. Transmitters operated between 173.605 and 173.977 MHz with each tag identified by a combination of frequency and pulse rate. The smallest tags (PIP type, mass = 1.4 g, dimensions = 18 mm × 8 mm) and intermediate sized tags were (TW-4 type, 2.1 g, 20 mm × 8 mm; 7.8 g, 30 × 15 mm) both had a trailing whip antenna. Prior to use these transmitters were activated and coated in a physiocompatible sealant (medical grade silicone elastomer PN400021, Applied Silicone Corporation). The large radio-transmitters (TW-3 type, 14.5 g, 60 mm × 15 mm) were sealed in a polycarbonate tube rounded at both ends with an internal coil antenna. The smallest and largest radio-transmitters, at a depth of 1 m in Slapton Ley, had ranges of, approximately 150 and 350 m, respectively, across open water, using a Mariner M57 receiver and three-element Yagi antenna.

Fish were anaesthetised on site (100 mg l^{-1} neutralised MS-222), placed in dorsal recumbency in a V-shaped surgical table and the gills irrigated with a flow of anaesthetic water (60 mg l^{-1} MS-222). The radio-transmitter was inserted into the body cavity through a small mid-ventral incision, just anterior to the pelvic girdle. A few scales had previously been removed from this area to accommodate the length of the incision. With the three smallest transmitters the antenna was passed through the ventro-lateral body wall, using a hypodermic needle guided over a needle shield. The incision was closed with two or three dissolvable sutures made of braided Polyglactin. The operation took 8–10 min. The animal was also given an intraperitoneal injection of a broad spectrum antibiotic (enrofloxacin, Baytril 2.5% injection) at a dose of 0.4 ml kg^{-1}, and the fork length and mass were measured. The transmitter/body mass ratio in air (average ratio = 1.2%) was maintained below the recommended maximum of 2% (Winter, 1996). After tagging, fish were allowed to recover in aerated lake water, and were then transferred to keep nets in the lake for several hours to allow further recovery before being released.

Ideally tagged fish would have been caught at several locations around the lake and released at those sites, in order to facilitate exposure to a range of environmental conditions. Although this was the case for pike, most perch were caught within the Lower Ley in the vicinity of Slapton Bridge and the channel that connects the Lower and Higher basins (Fig. 1). Therefore, a group of six perch were re-located to a site 500 m south west of this bridge, so ensuring that the perch were not just tracked in one area, where the Lower and Higher Leys meet, and that any movement into the neutral Higher Ley associated with changing water quality would be clear. Where possible the perch were released together with a large group of untagged conspecifics to reduce the risk of predation.

2.3.2. Data collection

The main period of data collection began 10 days after the last fish was tagged to allow the fish to recover and return to normal behaviour. The bulk of the data was collected over two periods of tracking (1–10 July and 5–13 August 2002), before and during the cyanobacterial phytoplankton bloom (and changes in water pH) in the Lower Ley. In both periods tracking was stratified over three time periods (British summer time): dawn (04:30–10:00), day (10:00–16:00) and dusk (16:00–21:30). Tracking during the hours of darkness was not performed for safety reasons. During both tracking periods four repetitions of each of the three time intervals were carried out and during each time interval as many fish as possible were located.

Fish were located by manoeuvring a 3 m boat, with tracking equipment, around the edge (approximately 50 m from the bank) of the Lower Ley, stopping every 100 m to scan for tagged fish in all directions. In consecutive tracking sessions a different path around the Lower Ley was taken. Preliminary data showed that most fish remained within 150 m of the shore, making this an efficient technique of locating most fish; however, additional transects were included further offshore to locate fish where time permitted. Tracking was also conducted as far into the Higher Ley as possible. A VHF radio-receiver (Mariner 57; Mariner Radar Ltd., Lowestoft, UK) and a directional three-element Yagi antenna, fixed to a rotating pole, were used to locate transmitters. Radio fixes were obtained by moving slowly towards the fish and adjusting gain and antenna direction until an omnidirectional signal was obtained from above or close to the transmitter. A hand-held global positioning system (Garmin GPS 12) together with landmarks were used to record fish positions on a map to within approximately 10 m. This precision was regarded as adequate for the purposes of the study. Four tagged perch were never found after release despite scanning the three tributaries to Slapton Ley and the Higher Ley on foot.

2.3.3. Water quality measurements

During every tracking session the water quality characteristics of the Lower and Higher Leys were recorded at several sites (Fig. 1). Water temperature and dissolved oxygen concentration were measured using a YSI 80 electrode, and water pH using a hand held meter (Hanna HI 8710) and low conductivity electrode (Russell KCET/11). All measurements were made approximately 0.5 m below the surface. These water quality measurements were also made at each radio-tagged fish location, so as to record the environmental conditions experienced by the fish.

2.3.4. Data analysis and statistics

The tracking data was visualised using Ranges 6 software (version 1.04, Anatrack Ltd.), separated by species and tracking period, combining locations from all individuals. Water quality parameters between the two tracking periods were compared using *t*-tests, with $P \leq 0.05$ as the level of significance.

3. Results

3.1. Laboratory experiments—chronic effects of high pH

Fig. 2 shows the Na^+ fluxes of the control fish, which remained in dechlorinated Exeter tap water at circumneutral pH (average = 7.40) throughout the experiment. Initially the control fish were in positive net Na^+ balance (J_{Na}^{net}), but this dropped significantly until it became indistinguishable from zero (i.e. net Na^+ balance) from day 3 onwards. Associated with this were significant reductions in Na^+ influx (J_{Na}^{in}) and Na^+ efflux (J_{Na}^{out}). In the experimental group (Fig. 2), fish were initially in positive net Na^+ balance at the pre-exposure time-point, but there was a significant and immediate 65% reduction in J_{Na}^{in} following exposure to high pH in the absence of any change in efflux, which resulted in J_{Na}^{net} becoming highly negative at time 0 h. By 8 h J_{Na}^{in} was not significantly different from the pre-exposure value and had fully recovered by 72 h. J_{Na}^{net} only partially recovered and remained slightly negative throughout the 7 days. J_{Na}^{out} was significantly stimulated relative to the pre-exposure value after 7 days of high pH, but was otherwise unaffected.

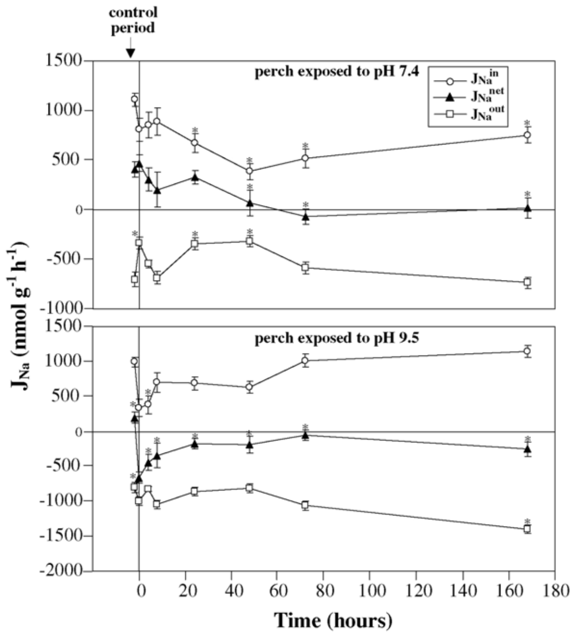

Fig. 2. Sodium fluxes in perch during the chronic high pH experiment. The control period is when both groups were in pH 7.4 water at the start of the experiment. Fish that remained in pH 7.4 water are in the upper panel and those exposed to pH 9.5 water are in the lower panel. Values are mean ± standard error. Asterisks (*) indicate a significant difference from the corresponding control period value.

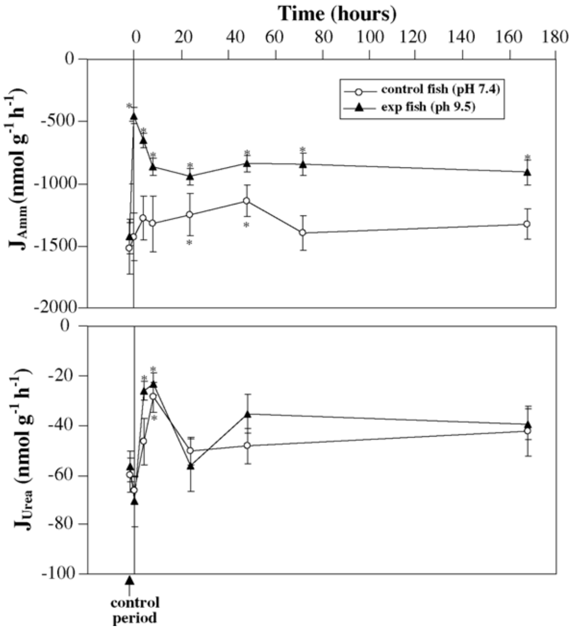

Fig. 3. Ammonia (upper panel) and urea (lower panel) excretion rates in perch during the chronic high pH exposure. The control period is when both groups were in pH 7.4 water at the start of the experiment. Values are mean ± standard error. Asterisks (*) indicate a significant difference from the corresponding control period value.

Ammonia excretion rates (J_{Amm}) for control fish varied somewhat with significant decreases of 20–30% at 24 and 48 h, but other values were not significantly different from the pre-exposure period (Fig. 3). There was an immediate and significant 68% decrease in J_{Amm} of experimental fish following exposure to alkaline water (Fig. 3). A partial recovery occurred but values remained 30–40% lower than the pre-exposure period from 8 h onwards. In control and experimental groups the urea excretion rates after 8 h were significantly lower than during the pre-exposure period (by 52 and 59%, respectively). Subsequently, urea excretion rates in both groups were not significantly different from the pre-exposure period.

3.2. Field sampling and physiological measurements

Fish sampled directly from the Lower Ley (perch and roach) had almost three-fold greater plasma total ammonia concentrations compared to fish sampled directly from the Higher Ley (perch) and fish held in Exeter dechlorinated tap water for 48 h (perch and roach) (Fig. 4). The plasma protein concentration in fish sampled directly from the Lower and Higher Leys were significantly higher (by about 40%) than those held in Exeter dechlorinated tap water (Fig. 4), but there was no difference between fish from the Higher and Lower Leys. There were no differences in the plasma Na^+ and Cl^- concentrations between the different groups of fish. Average plasma Na^+ and Cl^- was 134.9 ± 3.7 and $119.7 \pm 4.5\,\mathrm{mmol\,l^{-1}}$, respectively for perch and 132.7 ± 11.2 and $118.4 \pm 10.7\,\mathrm{mmol\,l^{-1}}$, respectively for roach.

3.3. Radio-tracking study—movements of fish within Slapton Ley

Average temperature, oxygen concentration and pH of the Lower Ley were all significantly higher in the second radio-tracking period compared to the first, with the average pH during the second period slightly exceeding the pH (9.50) used in the laboratory chronic

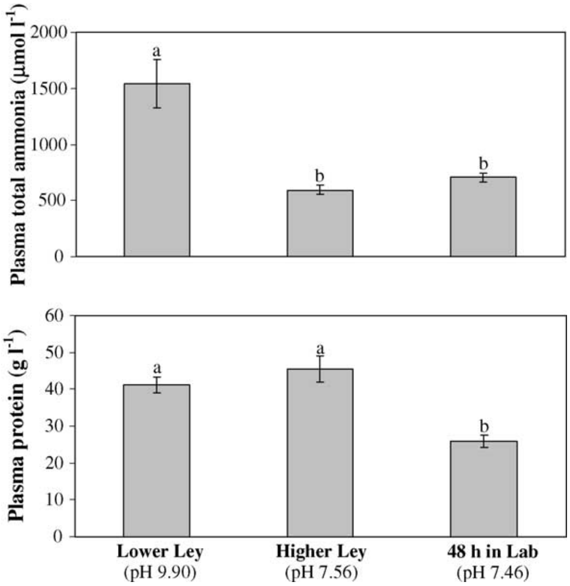

Fig. 4. Plasma ammonia and protein concentrations in fish from the Lower and Higher Leys and 48 h after return to the laboratory. Values are mean ± standard error, $n = 11, 10, 18$, respectively, for plasma ammonia and $n = 9, 5, 15$, respectively, for protein. Groups with different letters indicate a significant difference.

high pH experiment (Table 1). Only the temperature of the Higher Ley was significantly greater in the second tracking period compared to the first.

Although statistical comparisons of the locations of individual fish by home range analyses were not possible, due to the limited number of fixes for each fish (Kenward, 1992), it does not appear that pike or perch were utilising different parts of the lake during the two tracking periods (Fig. 5). In the first weeks after release, several tagged fish, especially perch, moved substantially (over 1000 m) within the Lower Ley from the sites of release, so that by the time of the first (pre-bloom) tracking period, they were relatively widely dispersed (Fig. 5). The behaviour of the fish during the two tracking periods was quite variable with some fish remaining fairly stationary during both the periods, some more mobile but remaining in similar areas during both periods, and some showing more movement within and between different areas during the two tracking periods. No movement of fish into the Higher Ley was recorded in either tracking period. There was also no clear diel pattern of fish movement.

The water quality at the sites where the fish were found was very similar to the averages for the Lower Ley (Fig. 6). There were a few pike locations with lower temperatures, oxygen concentrations and pHs than the average for the Lower Ley. Two pike were responsible for these points, as they both remained in the vicinity of Slapton Bridge during both tracking periods, where the water enters from the Higher Ley, which is shaded and does not generally contain algae blooms.

4. Discussion

4.1. Laboratory experiments—chronic effects of high pH

Control fish displayed significant changes in Na^+ and ammonia fluxes over time (Figs. 2 and 3). One would expect unstressed resting fish to be close to net Na^+ balance, but fish in both the control and high pH groups were in positive net Na^+ balance at the start of the experiment. This indicates the fish may not have recovered fully from anaesthesia and handling 24 h prior to the start of the experiment. When fish are acutely stressed an increase in intralamellar blood pressure

Table 1
Results of the water quality analysis of Slapton Ley (from seven sampling sites, see Fig. 1) during the two tracking periods, values are mean ± standard error

	Lower Ley		Higher Ley	
	First tracking period ($n = 70$)	Second tracking period ($n = 57$)	First tracking period ($n = 4$)	Second tracking period ($n = 4$)
Temperature (°C)	17.0 ± 0.1	20.2 ± 0.3*	15.1 ± 0.4	16.8 ± 0.5*
Oxygen (% saturation)	94.3 ± 2.3	134.3 ± 4.3*	59.6 ± 5.2	72.9 ± 8.8
Oxygen (mg l^{-1})	9.1 ± 0.2	11.9 ± 0.3*	6.0 ± 0.4	7.0 ± 0.6
pH	8.3 ± 0.1	9.9 ± 0.1*	7.2 ± 0.1	7.4 ± 0.2

Asterisks (*) indicate a significant difference from the first tracking period.

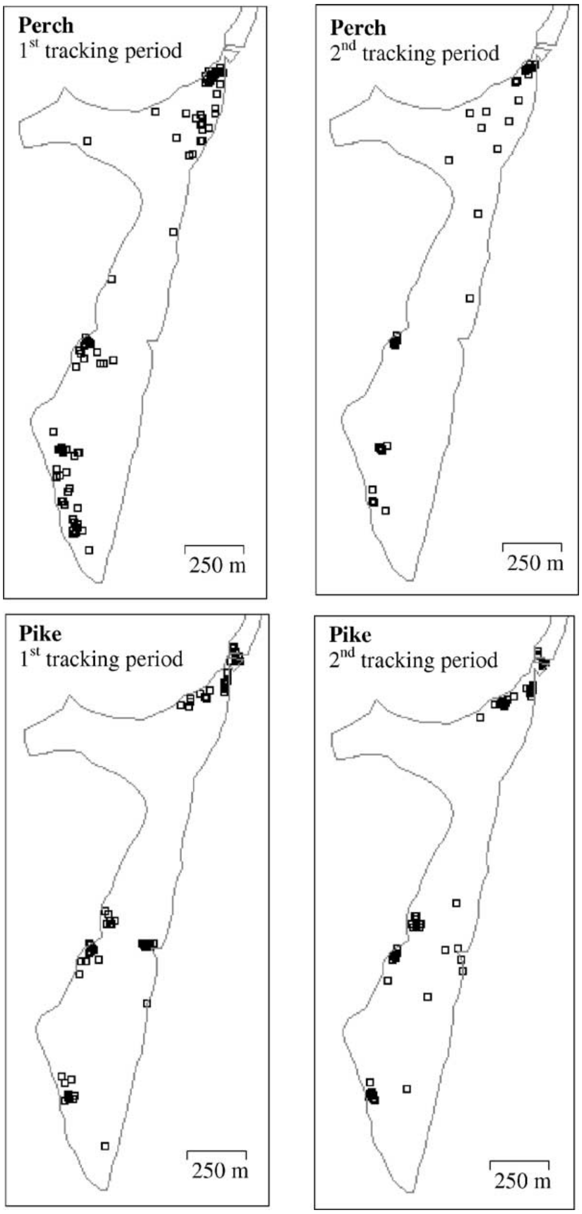

Fig. 5. Distribution of all perch ($n = 13$ for first tracking period, $n = 9$ for second tracking period) and pike ($n = 9$ for both the periods) in the Lower Ley during the two tracking periods (1st–10th July 2002 and 5th–13th August 2002). Open squares represent the locations of individual fish during each tracking period, where squares overlap there are solid blocks. As many fish as possible were located during dawn, day and dusk time periods.

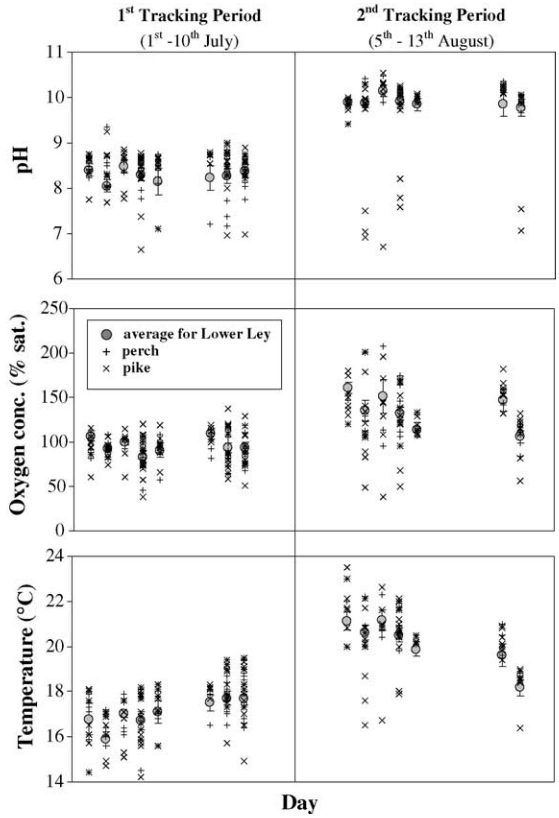

Fig. 6. The pH, oxygen and temperature of the lake water where individual perch and pike were located during the two tracking periods. The mean values ± standard error for the Lower Ley on each tracking day are also shown.

and diffusive ion loss occurs (McDonald et al., 1991; Gonzalez and McDonald, 1992; McDonald and Milligan, 1997; Wendelaar Bonga, 1997; Perry and Bernier, 1999). Therefore, following handling and anaesthesia, fish may have suffered sufficient diffusive ion loss to require a compensatory upregulation of Na^+ uptake and subsequently a positive net flux at the start of the experiment. Excretion of ammonia in starved fish represents the normal protein and nucleic acid turnover required for body maintenance (Wood, 2001). Therefore the slight but significant decreases in ammonia excretion in the control fish at 24 and 48 h may be due to changes in metabolic rate.

The immediate inhibitory effect of high pH water on Na^+ uptake, but not its complete recovery by 72 h, was similar to that seen in rainbow trout by other workers (Wilkie and Wood, 1994; Wilkie et al., 1999; Laurent et al., 2000). This suggests that, unlike trout, perch do not employ permanent lowering of Na^+/ H^+ exchange rate as a way to offset chronic alkalosis. All the net Na^+ flux values were significantly different from the pre-exposure value in the experimental fish, but this was mainly due to the highly positive net Na^+ flux in the pre-exposure period rather than Na^+ imbalance later on. By 48 and 72 h the net Na^+ flux was close to zero and so fish were near to exact net Na^+ balance, but after 1 week at pH 9.50 the net Na^+ flux was somewhat negative ($-252\,nmol\,g^{-1}\,h^{-1}$) associated with a significantly increased Na^+ efflux. It is, therefore, not clear if there is a genuine long-term effect of high pH water on Na^+ balance in these fish.

Ammonia excretion (Fig. 3) was initially inhibited by 68%, which is within the 40–100% range documented for rainbow trout exposed to pH 9.5 water (Wright and Wood, 1985; Wilkie and Wood, 1991, 1994, 1995; Wilkie et al., 1996; Laurent et al., 2000). Partial recovery of ammonia excretion occurred within the first 8 h but thereafter ammonia excretion remained 40–60% lower than the control fish. To avoid ammonia toxicity these fish must have made adjustments to cope with the shortfall, such as a reduction in endogenously produced ammonia (McGeer and Eddy, 1998; Wilson et al., 1998), conversion of excess ammonia to less toxic compounds such as glutamine (Arillo et al., 1981; Wicks and Randall, 2002), and storage in tissues such as white muscle (Wilkie and Wood, 1995; Wilkie et al., 1996). Urea is thought to be a potential detoxification product in rainbow trout (Wilkie and Wood, 1991; Wilkie et al., 1996), but this does not appear to be relevant in perch as there was no increase in urea excretion associated with exposure to high pH water.

4.2. Field sampling and physiological measurements

Examination of blood parameters in fish taken directly from Slapton Ley was, due to logistical constraints, a limited analysis of perch and roach samples. Nevertheless, some important patterns in the wild fish from a hyper-eutrophic, seasonally alkaline lake were revealed. Slapton Ley serves as a useful natural experiment as it consists of two connected basins that share the same primary water supply, and yet experience substantially different water pHs during the summer months. Fish caught in the alkaline Lower Ley (pH 9.90) had plasma total ammonia concentrations nearly three-fold greater than fish from the neutral Higher Ley (pH 7.56), or fish held for a further 48 h in neutral pH waters in the laboratory. These results indicate that fish in the Lower Ley experienced substantial disruption to their ammonia excretion compared to the fish in more neutral waters. Elevated body ammonia levels are known to be toxic in fish (Ip et al., 2001) and are likely to cause some disruption to the normal physiology and activity of the wild fish in Slapton Ley. In particular, there may be detrimental effects on muscle function and swimming performance, as has been seen in salmonids (Beaumont et al., 1995, 2003; Shingles et al., 2001; Wicks et al., 2002).

Rapid changes in plasma protein concentration are usually considered to reflect shifts in water between the plasma volume and other compartments (McDonald et al., 1980). Branchial ion loss will cause a reduction in plasma ions and an osmotic shift of fluid from the plasma to the tissues, thus concentrating proteins in the smaller plasma volume (Milligan and Wood, 1982). Plasma protein analysis in this study only revealed significant differences between field and laboratory sampled fish and not between fish from the different pH locations within the Slapton Ley (Fig. 4). This suggests that the electric fishing process had an effect on plasma protein but that high pH did not. It is likely that diuresis associated with handling and forced exercise (Hickman and Trump, 1969; Wood and Randall, 1973) and the resultant reduction in plasma volume is the cause of the post-capture elevations in plasma protein levels in the field.

4.3. Radio-tracking study—movements of fish within Slapton Ley

Overall this radio-tracking study answered one important question, of whether fish try to avoid the alkaline conditions in a hyper-eutrophic lake during summer phytoplankton blooms. The answer to this question appears to be no, at least with respect to adult pike and perch. It is unclear whether: (a) these fish were simply not affected by the alkaline conditions in the Lower Ley; (b) the advantages of staying (e.g. better foraging) were greater than seeking refuge in more neutral pH water; (c) fish were "sitting out" the adverse conditions; or (d) that they were unable to find refuge. The latter seems unlikely as perch and pike are known to migrate into the Higher Ley to spawn and perch are thought to move freely throughout the two Leys (Craig, 1974; Bregazzi, 1978; Bregazzi and Kennedy, 1980). At the time of the present study the two streams that enter the Lower Ley were barely a trickle and were probably not available for use by fish. Fish moving between the two basins must pass through a narrow (10 m) relatively unvegetated channel making it a good ambushing site for predatory pike. Several tagged pike remained in the area of this channel during both tracking periods. Therefore movement into the Higher Ley would be accompanied by increased predation risks, which might outweigh the advantages of using the Higher Ley as a refuge from poor water quality. During spawning migrations into the Higher Ley fish move en masse, which presumably provides protection when travelling through this exposed channel.

It seems quite remarkable that the fish in Slapton Ley are able to withstand such high pHs, including values of up to pH 10.54 at some locations where radio-tagged fish were found. In enclosure experiments perch survived pH 10 but died in pH 11 water (Beklioglu and Moss, 1995), suggesting that pHs experienced by perch during August 2003 in the Lower Ley were near the upper limit for survival. Several laboratory studies have shown that rainbow trout can survive high pH water for a considerable length of time (15 days at pH 10 and 28 days at pH 9.5) due to adjustments to branchial ion movements, acid–base balance, and ammonia excretion and production (Wilkie et al., 1996; Wilson et al., 1998; Ip et al., 2001), which may be the same mechanisms employed by wild fish in Slapton Ley.

The lack of specific patterns of movement by tagged pike and perch during the two tracking periods is not surprising, as this has been recorded previously. Pike may adopt a home range for an extended period (up to several years), or display frequent habitat shifts (Mann, 1980; Chapman and Mackay, 1984a, 1984b; Eklöv, 1997; Jepsen et al., 2001). Similarly, large perch have been shown to have either no specific home range or a consistent site fidelity (Eklöv, 1997; Zamora and Moreno-Amich, 2002). Previous mark-recapture studies of these species in Slapton have also revealed no particular patterns of movement (Craig, 1974; Bregazzi, 1978; Bregazzi and Kennedy, 1980). Detailed analysis of fish movements was limited by the small number of fixes (1–12) recorded for each fish over variable time intervals during each tracking period. This can be mainly attributed to the size of Slapton Ley relative to the range of the radio-transmitters used; the minimum transmitter range was, approximately 150 m and the Lower Ley alone covers an area of 77 ha (770,000 m^2). With limited human resources it was impossible to locate every fish as regularly as desired. Nevertheless, this study produced some useful information on the movement of pike and perch in a hyper-eutrophic lake before and during a phytoplankton bloom.

Based on a combination of chronic laboratory experiments with perch, and field measurements of physiology (perch and roach) and behaviour (perch and pike) it seems that during prolonged alkaline exposure in the summer months at Slapton Ley fish experience no significant disturbances to Na^+ balance but suffer from sustained reduction in ammonia excretion and elevated plasma ammonia levels. Radio-tracking in the field revealed that adult pike and perch did not move out of the Lower Ley during a phytoplankton bloom despite the presence of potentially harmful alkaline conditions (up to pH 10.54). The reason for this is unclear, but it seems likely that the advantages (i.e. better foraging, less predation) of remaining and withstanding the conditions in the Lower Ley were greater than moving into the less productive but pH neutral Higher Ley. This study is a unique combination of laboratory and field techniques in order to investigate the potential effects of eutrophication (particularly high pH) on non-salmonid species, and may serve as a useful example for an integrated approach to investigate the effect of toxicants on fish in general.

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References

Arillo, A., Margiocco, C., Melodia, F., Mensi, P., Schenone, G., 1981. Ammonia toxicity mechanism in fish: studies on rainbow trout (*Samo gairdneri* Rich.). Ecotoxicol. Environ. Safety 5, 316–328.

Beaumont, M.W., Butler, P.J., Taylor, E.W., 1995. Plasma ammonia concentration in brown trout in soft acidic water and its relationship to decreased swimming performance. J. Exp. Biol. 198, 2213–2220.

Beaumont, M.W., Butler, P.J., Taylor, E.W., 2003. Exposure of brown trout *Salmo trutta* to a sub-lethal concentration of copper in soft acidic water: effects upon gas exchange and ammonia accumulation. J. Exp. Biol. 206, 153–162.

Beklioglu, M., Moss, B., 1995. The impact of pH on interactions among phytoplankton algae, zooplankton and perch (*Perca fluviatilis*) in a shallow, fertile lake. Freshwater Biol. 33, 497–509.

Bregazzi, P.R., 1978. The biology and management of pike (*Esox lucius*) and perch (*Perca fluviatilis*) in Slapton Ley, Devon. University of Exeter, Ph.D. thesis.

Bregazzi, P.R., Kennedy, C.R., 1980. The biology of pike, *Esox lucius* L., in a southern eutrophic lake. J. Fish Biol. 17, 91–112.

Brett, J.R., Zala, C.A., 1975. Daily pattern of nitrogen excretion and oxygen consumption of sockeye salmon (*Oncorhynchus nerka*) under controlled conditions. J. Fish. Res. Board Can. 32, 2479–2486.

Carline, R.F., DeWalle, D.R., Sharpe, W.E., Dempsey, B.A., Gagen, C.J., Swistock, B., 1992. Water chemistry and fish community responses to episodic stream acidification in Pennsylvania. USA. Environ. Pollut. 78, 45–48.

Chapman, C.A., Mackay, W.C., 1984a. Versatility in habitat use by a top aquatic predator *Esox lucius* L. J. Fish Biol. 25, 109–115.

Chapman, C.A., Mackay, W.C., 1984b. Direct observation of habitat utilization by northern pike. Copeia, 225–228.

Craig, J.F., 1974. Population dynamics of perch, *Perca fluviatilis* L. in Slapton Ley, Devon. I. Trapping behaviour, reproduction, migration, population estimates, mortality and food. Freshwater Biol. 4, 417–431.

Craig, J.F., 1987. The Biology of the Perch and Related Fish. Academic Press, Portland.

Eklöv, P., 1997. Effects of habitat complexity and prey abundance on the spatial and temporal distributions of perch (*Perca fluviatilis*) and pike (*Esox lucius*). Can. J. Fish. Aquat. Sci. 54, 1520–1531.

Gonzalez, R.J., McDonald, D.G., 1992. The relationship between oxygen consumption and ion loss in a freshwater fish. J. Exp. Biol. 163, 317–332.

Goss, G.G., Perry, S.F., Wood, C.M., Laurent, P., 1992. Mechanisms of ion and acid–base regulation at the gills of freshwater fish. J. Exp. Zool. 263, 143–159.

Hickman, C.P., Trump, B., 1969. The kidney. In: Hoar, W.S., Randall, D.J. (Eds.), Fish Physiology, vol. 1. Academic Press, London, pp. 91–240.

Ip, Y.K., Chew, S.F., Randall, D.J., 2001. Ammonia toxicity, tolerance and excretion. Fish Physiology, vol. 20. Academic Press, London, 109–148.

Jepsen, N., Beck, S., Skov, C., Koed, A., 2001. Behaviour of pike (*Esox lucius* L.) >50 cm in a turbid reservoir and in a clearwater lake. Ecol. Freshwater Fish 10, 26–34.

Jepsen, N., Koed, A., Thorstad, E.B., Baras, E., 2002. Surgical implantation of telemetry transmitters in fish: how much have we learned? Hydrobiology 483, 239–248.

Johnes, P.J., Heathwaite, A.L., 1997. Modelling the impact of land use change on water quality in agricultural catchments. Hydrol. Processes 11, 269–286.

Kenward, R.E., 1992. Quantity versus quality: programmed collection and analysis of radio-tracking data. In: Priede, I.G., Swift, S.M. (Eds.), Wildlife Telemetry. Ellis Horwood Ltd., Chichester, pp. 231–246.

Laurent, P., Wilkie, M.P., Chevalier, C., Wood, C.M., 2000. The effect of highly alkaline water (pH 9.5) on the morphology and morphometry of chloride cells and pavement cells in the gills of the freshwater rainbow trout: relationship to ionic transport and ammonia excretion. Can. J. Zool. 78, 307–319.

Lowry, O.H., Roseburg, N.J., Farr, A.L., Randall, R.J., 1951. Protein measurement with the folin phenol reagent. J. Biol. Chem. 193, 265–275.

Lucas, M.C., Baras, E., 2000. Methods for studying spatial behaviour of freshwater fishes in the natural environment. Fish Fisheries 1, 283–316.

Mann, R.H.K., 1980. The numbers and production of pike (*Esox lucius*) in two Dorset rivers. J. Fish Biol. 49, 899–915.

McDonald, D.G., Wood, C.M., 1981. Branchial and renal acid and ion fluxes in the rainbow trout, *Salmo gairdneri*, at low environmental pH. J. Exp. Biol. 93, 101–118.

McDonald, G., Milligan, L., 1997. Ionic, osmotic and acid–base regulation in stress. In: Iwama, G.K., Pickering, A.D., Sumpter, J.P., Schrek, C.B. (Eds.), Fish Stress and Health in Aquaculture. Cambridge University Press, Cambridge, pp. 119–144.

McDonald, D.G., Hobe, H., Wood, C.M., 1980. The influence of calcium on the physiological responses of the rainbow trout, *Salmo gairdneri*, to low environmental pH. J. Exp. Biol. 88, 109–131.

McDonald, D.G., Cavdek, V., Ellis, R., 1991. Gill design in freshwater fishes: interrelationships among gas-exchange, ion regulation, and acid–base regulation. Physiol. Zool. 64, 103–123.

McGeer, J.C., Eddy, F.B., 1998. Ionic regulation and nitrogenous excretion in rainbow trout exposed to buffered and unbuffered freshwater of pH 10.5. Physiol. Zool. 71, 179–190.

Miller, G.L., 1959. Protein determination for large numbers of samples. Anal. Chem. 31, 964.

Milligan, C.L., Wood, C.M., 1982. Disturbances in haematology, fluid volume distribution and circulatory function associated with low environmental pH in the rainbow trout, *Salmo gairdneri*. J. Exp. Biol. 99, 397–415.

Morey, C.R., 1976. The natural history of Slapton Ley nature reserve. IX. The morphology and history of the lake basins. Field Stud. 4, 353–368.

O'Sullivan, P.E., 1994. The natural history of Slapton Ley nature reserve XXI: the palaeolimnology of the uppermost sediments of the Lower Ley, with interpretations based on ^{210}Pb dating and the historical record. Field Stud. 8, 403–449.

O'Sullivan, P.E., Heathwaite, A.L., Appleby, P.G., Brookfield, D., Crick, M.W., Moscrop, C., Mulder, T.B., Vernon, N.J., Wilmshurst, J.M., 1991. Palaeolimnology of Slapton Ley, Devon UK. Hydrobiology 214, 115–124.

Perry, S.F., Bernier, N.J., 1999. The acute humoral adrenergic stress response in fish: facts and fiction. Aquaculture 177, 285–295.

Priede, I.G., Swift, S.M., 1992. Wildlife Telemetry, Remote Monitoring and Tracking of Animals. Ellis Horwood Ltd., London.

Rahmatullah, M., Boyde, T.R.C., 1980. Improvements in the determination of urea using diacetyl monoxime; methods with and without deproteinisation. Clin. Chem. Acta 107, 3–9.

Salama, A., Morgan, I.J., Wood, C.M., 1999. The linkage between Na^+ uptake and ammonia excretion in rainbow trout: kinetic analysis, the effects of $(NH_4)_2SO_4$ and NH_4HCO_3 infusion and the influence of gill boundary layer pH. J. Exp. Biol. 202, 697–709.

Scott, D.M., 2003. The physiology and behaviour of fish from a freshwater, eutrophic lake, Slapton Ley, Devon. University of Exeter, Ph.D. thesis.

Serfay, J.E., Harrell, R.M., 1993. Behavioural response of fishes to increasing pH and dissolved oxygen: field and laboratory observations. Freshwater Biol. 30, 53–61.

Shingles, A., McKenzie, D.J., Taylor, E.W., Moretti, A., Butler, P.J., Ceradini, S., 2001. Effects of sublethal ammonia exposure on swimming performance in rainbow trout (*Oncorhynchus mykiss*). J. Exp. Biol. 204, 2691–2698.

Van Vlymen, C.D., 1980. The water balance, physico-chemical environment and phytoplankton studies of Slapton Ley, Devon. University of Exeter, Ph.D. thesis.

Verdouw, H., Van Echteld, C.J.A., Dekkers, E.M.J., 1978. Ammonia determination based on indophenol formation with sodium salicylate. Water Res. 12, 399–402.

Wendelaar Bonga, S.E., 1997. The stress response in fish. Physiol. Rev. 77, 591–625.

Wicks, B.J., Randall, D.J., 2002. The effect of sub-lethal ammonia exposure on fed and unfed rainbow trout: the role of glutamine in regulation of ammonia. Comp. Biochem. Physiol. A 132, 275–285.

Wicks, B.J., Joensen, R., Tang, Q., Randall, D.J., 2002. Swimming and ammonia toxicity in salmonids: the effect of sub lethal ammonia exposure on the swimming performance of coho salmon and the acute toxicity of ammonia in swimming and resting rainbow trout. Aquat. Toxicol. 59, 55–69.

Wilkie, M.P., Wood, C.M., 1991. Nitrogenous waste excretion, acid–base regulation, and ionoregulation in rainbow trout (*Oncorhynchus mykiss*) exposed to extremely alkaline water. Physiol. Zool. 64, 1069–1086.

Wilkie, M.P., Wood, C.M., 1994. The effects of extremely alkaline water (pH 9.5) on rainbow trout gill function and morphology. J. Fish Biol. 45, 87–98.

Wilkie, M.P., Wood, C.M., 1995. Recovery from high pH exposure in the rainbow trout—white muscle ammonia storage, ammonia washout, and the restoration of blood chemistry. Physiol. Zool. 68, 379–401.

Wilkie, M.P., Wright, P., Iwama, G.K., Wood, C.M., 1993. The physiological responses of the Lahontan cutthroat trout (*Oncorhynchus clarki henshawi*), a resident of highly alkaline pyramid lake (pH 9.4), to challenge at pH 10. J. Exp. Biol. 175, 173–194.

Wilkie, M.P., Simmons, H.E., Wood, C.M., 1996. Physiological adaptations of rainbow trout to chronically elevated water pH (pH 9.5). J. Exp. Zool. 274, 1–14.

Wilkie, M.P., Laurent, P., Wood, C.M., 1999. The physiological basis for altered Na^+ and Cl^- movements across the gills of rainbow trout (*Oncorhynchus mykiss*) in alkaline (pH 9.5) water. Physiol. Biochem. Zool. 72, 360–368.

Wilson, J.M., Iwata, K., Iwama, G.K., Randall, D.J., 1998. Inhibition of ammonia excretion and production in rainbow trout during severe alkaline exposure. Comp. Biochem. Physiol. 121, 99–109.

Wilson, R.W., Wright, P.M., Munger, S., Wood, C.M., 1994. Ammonia excretion in fresh water rainbow trout (*Oncorhynchus mykiss*) and the importance of gill boundary-layer acidification: lack of evidence for Na^+/NH_4^+ exchange. J. Exp. Biol. 191, 37–58.

Winter, J.D., 1996. Advances in underwater biotelemetry. In: Murphy, B.R., Willis, D.W. (Eds.), Fisheries Techniques. American Fisheries Society, Maryland, pp. 555–590.

Wood, C.M., 2001. Influence of feeding, exercise, and temperature on nitrogen metabolism and excretion. In: Wright, P.A., Anderson, P.M. (Eds.), Fish Physiology, vol. 20. Academic Press, London, pp. 201–238.

Wood, C.M., Randall, D.J., 1973. The influence of swimming activity on water balance in the rainbow trout (*Salmo gairdneri*). J. Comp. Physiol. 82, 257–276.

Wright, P.A., Wood, C.M., 1985. An analysis of branchial ammonia excretion in the fresh water rainbow trout—effects of environmental pH change and sodium uptake blockade. J. Exp. Biol. 114, 329–353.

Wright, P.A., Randall, D.J., Perry, S.F., 1989. Fish gill water boundary layer—a site of linkage between carbon dioxide and ammonia excretion. J. Comp. Physiol. B 158, 627–635.

Yesaki, T.Y., Iwama, G.K., 1992. Survival, acid–base regulation, ion regulation, and ammonia excretion in rainbow trout in highly alkaline hard water. Physiol. Zool. 65, 763–787.

Zall, D.M., Fisher, D., Garner, M.Q., 1956. Photometric determination of chlorides in water. Anal. Chem. 28, 1665–1668.

Zamora, L., Moreno-Amich, R., 2002. Quantifying the activity and movement of perch in a temperate lake by integrating acoustic telemetry and a geographic information system. Hydrobiology 483, 209–218.